Isaac Loc

locisaac1223@gmail.com | linkedin.com/in/isaac-loc | github.com/Isaac-Loc | isaac-loc.dev

Education

University at Buffalo Buffalo, NY

Bachelor of Science in Computer Science, Minor in Math *Expected May 2027*

- **Relevant Coursework:** Computer Organization, Algorithms and Complexity, Applied Probability and Statistics, Systems Programming, Data Structures and Algorithms, Object Oriented Programming, Software Engineering

Experience

Full Stack Software Engineer Remote

EZEsports – ezesports.org | github.com/EZ-Esports/website *Sept 2026 – Present*

- Diagnosed a production image outage (**5,040 Vercel transformations** vs a **5,000/month** limit) and eliminated gallery transformations with **WebP thumbnails** up to **39x smaller** served from **Supabase**.
- Closed a **privilege-escalation** path letting non-owner roles grant **Owner**, by fixing role-ordering logic in **TypeScript** and enforcing owner protection across **15 server-side checks** with new unit tests.
- Secured applicant data with **PDF resume uploads** validated **100% server-side** in a private **Supabase** bucket (**60-second signed URLs**) and on-request erasure of **100% of applicant data** via **PostgreSQL** triggers.

Freelance Software Engineer Remote

Self-employed *Sept 2026 – Present*

- Delivered a fast, low-maintenance static website by building it with **React** and **Vite**, shipping a **∼79 KB gzipped** JavaScript bundle, and hosting it on **Azure** with a custom domain and managed **HTTPS**.
- Automated releases across **2 environments** (production on Azure, staging on **Vercel**) with a **GitHub Actions** CI/CD pipeline that builds and deploys every push to **main**.
- Hardened the site against clickjacking and sniffing with **3 HTTP security headers** and **immutable 1-year caching** for built assets.

Embedded Software Engineer Buffalo, NY

University at Buffalo – CSE 302 *Aug 2025 – Dec 2025*

- Synchronized real-time video and image playback across a **50+ LED tile array** by developing **C++** firmware with **ESP-IDF** to coordinate **4 ESP32** microcontrollers and keep every tile in sync.
- Orchestrated live canvas manipulation across **10+ command types** without interrupting active playback, by implementing a C++ command interface backed by a real-time message queue.
- Connected the web frontend to the C++ render loop via **4 REST endpoints**, by building a **Python** canvas library behind a **Flask** API in weekly Agile sprints.

Projects

Project Management Tool | *React, PHP/Slim, MySQL* GitHub

- Delivered a platform used by **20+ members across 3+ teams**, by architecting a **React** frontend and **PHP/Slim** API with **68 endpoints** over a **17-table MySQL** database.
- Streamlined task tracking with a **6-stage Kanban** board and dependency graph in React, and enabled in-app file access via the **Google Drive API** with OAuth token refresh.

Bidit Auction Marketplace | *React, PHP, MySQL*

- Launched a peer-to-peer marketplace with **2 listing types** (live auctions and buy-it-now), by building a full-stack app with **React**, **PHP**, and **MySQL** for users, listings, and bids.

Leadership & Extracurricular

Brotherhood Outreach & Software Lead | *Theta Tau – Engineering Fraternity* May 2025 – Present

- Built an **LED wall** as software lead to advertise the organization at tabling events, using **C++**, **PlatformIO**, and **ESP32**s driving a **HUB75** LED layout, increasing tabling attraction by **25%**.

Technical Skills

Frameworks & Libraries: React, Next.js, Tailwind CSS, React Aria, Drizzle ORM, PHP/Slim, Flask, ESP-IDF
Databases: PostgreSQL, MySQL, Supabase
Tools & Platforms: Git, GitHub, Docker, Linux, SSH, Microsoft Azure (Static Web Apps), CI/CD (GitHub Actions), Vite, Node.js, Vercel, Vitest, CMake, PlatformIO, ESP32, HUB75, REST APIs, OAuth 2.0, DNS/HTTPS, Agile/Scrum, Claude Code, AI-Assisted Development
Languages: C/C++, Python, TypeScript, JavaScript, PHP, SQL, HTML/CSS, Java, ARM Assembly